

UNIVERSIDAD DE GUANAJUATO



Análisis de color

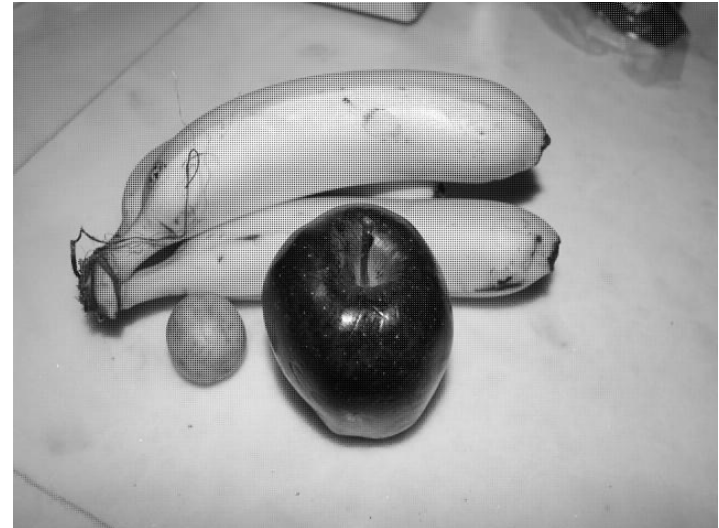
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Oficina: 217



$$\begin{bmatrix} B_{11} & G_{12} & B_{13} & G_{14} \\ G_{21} & R_{22} & G_{23} & R_{24} \\ B_{31} & G_{32} & B_{33} & G_{34} \\ G_{41} & R_{42} & G_{43} & R_{44} \end{bmatrix} = \begin{bmatrix} R_{11} & R_{12} & R_{13} & R_{14} \\ R_{21} & R_{22} & R_{23} & R_{24} \\ R_{31} & R_{32} & R_{33} & R_{34} \\ R_{41} & R_{42} & R_{43} & R_{44} \end{bmatrix} + \begin{bmatrix} G_{11} & G_{12} & G_{13} & G_{14} \\ G_{21} & G_{22} & G_{23} & G_{24} \\ G_{31} & G_{32} & G_{33} & G_{34} \\ G_{41} & G_{42} & G_{43} & G_{44} \end{bmatrix} + \begin{bmatrix} B_{11} & B_{12} & B_{13} & B_{14} \\ B_{21} & B_{22} & B_{23} & B_{24} \\ B_{31} & B_{32} & B_{33} & B_{34} \\ B_{41} & B_{42} & B_{43} & B_{44} \end{bmatrix}$$

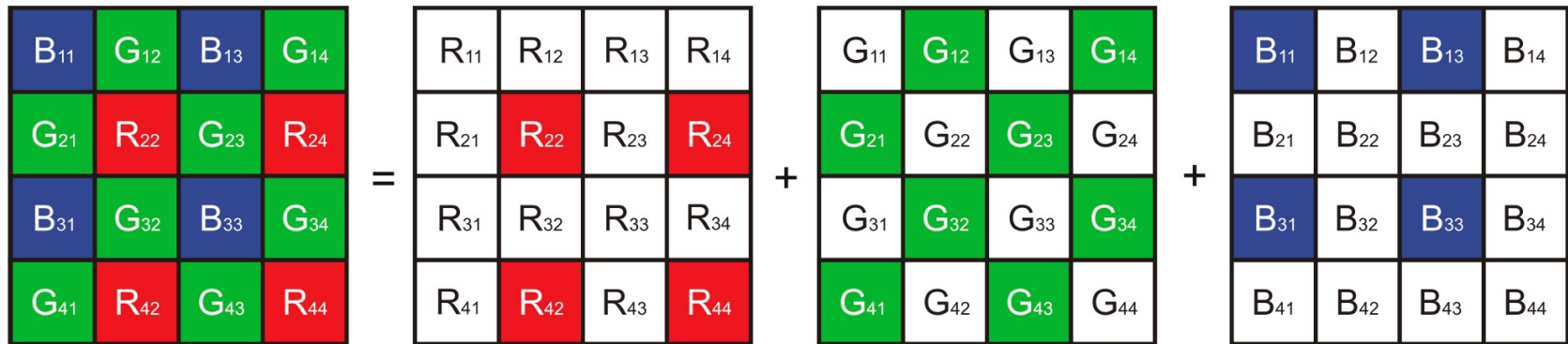


$$c(u, v) = \sum_{k=1}^M \sum_{l=1}^N c(u_k, v_l) h(|u - u_k|) h(|v - v_k|)$$

ANÁLISIS DEL COLOR / Demosaicado por próximos vecinos



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$$c(u, v) = \sum_{k=1}^M \sum_{l=1}^N c(u_k, v_l) h(|u - u_k|) h(|v - v_k|)$$

$$h(x) = \begin{cases} 0 & 0 \leq x \leq 0,5 \\ 1 & x > 0,5 \end{cases}$$

```
clc;
close all;
clear;

if ispc
    d = '\\';
else
    d = '/';
end

path_bd = pwd;
ind_d = find(path_bd==d);
path_bd = [path_bd(1:ind_d(end-1)) 'bd' d 'Color' d 'rggb' d];

fn_img = 'Frutas07r_rggb.jpg';

imgRGGB = imread([path_bd fn_img]);

[lv, lu] = size(imgRGGB);

idu = mod(ones(lv,1)*(1:lu),2)==1;
idv = mod((1:lv)'*ones(1,lu),2)==1;
idR = idu&idv;
idG1 = ~idu&idv;
idG2 = idu&~idv;
idB = ~idu&~idv;
```

```
imgRGB = zeros(lv, lu, 3, 'uint8');  
imgR = zeros(lv, lu, 'uint8');  
imgG = zeros(lv, lu, 'uint8');  
imgB = zeros(lv, lu, 'uint8');
```

```
imgR(idR) = imgRGGB(idR);  
imgR(idG1) = imgRGGB(idR);  
imgR(idG2) = imgRGGB(idR);  
imgR(idB) = imgRGGB(idR);
```

```
imgG(idR) = uint8((double(imgRGGB(idG1))+double(imgRGGB(idG2)))/2);  
imgG(idG1|idG2) = imgRGGB(idG1|idG2);  
imgG(idB) = uint8((double(imgRGGB(idG1))+double(imgRGGB(idG2)))/2);
```

```
imgB(idR) = imgRGGB(idB);  
imgB(idG1) = imgRGGB(idB);  
imgB(idG2) = imgRGGB(idB);  
imgB(idB) = imgRGGB(idB);
```

```
imgRGB(:, :, 1) = imgR;  
imgRGB(:, :, 2) = imgG;  
imgRGB(:, :, 3) = imgB;
```

B ₁₁	G ₁₂	B ₁₃	G ₁₄	B ₁₅	G ₁₆
G ₂₁	R ₂₂	G ₂₃	R ₂₄	G ₂₅	R ₂₆
B ₃₁	G ₃₂	B ₃₃	G ₃₄	B ₃₅	G ₃₆
G ₄₁	R ₄₂	G ₄₃	R ₄₄	G ₄₅	R ₄₆
B ₅₁	G ₅₂	B ₅₃	G ₅₄	B ₅₅	G ₅₆
G ₆₁	R ₆₂	G ₆₃	R ₆₄	G ₆₅	R ₆₆

$$c(u, v) = \sum_{k=1}^M \sum_{l=1}^N c(u_k, v_l) h(|u - u_k|) h(|v - v_k|)$$

$$h_R(x) = \left\{ \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}, \quad h_G(x) = \left\{ \frac{1}{2} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right\}, \quad h_B(x) = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \right\}.$$

```
lv2 = lv/2;  
lu2 = lu/2;  
imgRGB2 = zeros(lv2,lu2,3,'uint8');  
imgRGB2(1:(lv2*lu2)) = imgRGGB(idR);  
imgRGB2(lv2*lu2+(1:(lv2*lu2))) = uint8((double(imgRGGB(idG1))+double(imgRGGB(idG2)))/2);  
imgRGB2(2*lv2*lu2+(1:(lv2*lu2))) = imgRGGB(idB);
```


B ₁₁	G ₁₂	B ₁₃	G ₁₄	B ₁₅	G ₁₆
G ₂₁	R ₂₂	G ₂₃	R ₂₄	G ₂₅	R ₂₆
B ₃₁	G ₃₂	B ₃₃	G ₃₄	B ₃₅	G ₃₆
G ₄₁	R ₄₂	G ₄₃	R ₄₄	G ₄₅	R ₄₆
B ₅₁	G ₅₂	B ₅₃	G ₅₄	B ₅₅	G ₅₆
G ₆₁	R ₆₂	G ₆₃	R ₆₄	G ₆₅	R ₆₆

$$h_R(x) = \begin{cases} \frac{1}{2} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} & x \in x_G, \\ \frac{1}{4} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix} & x \in x_B. \end{cases}$$

$$h_G(x) = \begin{cases} \frac{1}{4} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} & x \in \{x_R, x_B\}. \end{cases}$$

$$c(u, v) = \sum_{k=1}^M \sum_{l=1}^N c(u_k, v_l) h(|u - u_k|) h(|v - v_k|)$$

$$h_B(x) = \begin{cases} \frac{1}{2} \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} & x \in x_G, \\ \frac{1}{4} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix} & x \in x_R. \end{cases}$$

```
M1 = [0 1 0; 0 0 0; 0 1 0];
M2 = [0 0 0; 1 0 1; 0 0 0];
M3 = [1 0 1; 0 0 0; 1 0 1];
M4 = [0 1 0; 1 0 1; 0 1 0];

imgRGB3 = zeros(lv, lu, 3, 'uint8');
for i1=2:lv-1
    for i2=2:lu-1
        switch bitor(mod(i1,2)*2,mod(i2,2))
            case 0 % Par(lv)-par(lu)
                aux = double(imgRGGB(i1-1:i1+1,i2-1:i2+1)).*M3;
                imgRGB3(i1, i2, 1) = uint8(sum(aux(:))/sum(M3(:)));
                aux = double(imgRGGB(i1-1:i1+1,i2-1:i2+1)).*M4;
                imgRGB3(i1, i2, 2) = uint8(sum(aux(:))/sum(M4(:)));
                imgRGB3(i1, i2, 3) = imgRGGB(i1, i2);
            case 1 % Par(lv)-impar(lu)
                aux = double(imgRGGB(i1-1:i1+1,i2-1:i2+1)).*M1;
                imgRGB3(i1, i2, 1) = uint8(sum(aux(:))/sum(M1(:)));
                imgRGB3(i1, i2, 2) = imgRGGB(i1,i2);
                aux = double(imgRGGB(i1-1:i1+1,i2-1:i2+1)).*M2;
                imgRGB3(i1, i2, 3) = uint8(sum(aux(:))/sum(M2(:)));
```

```
case 2 % Impar(lv)-par(lu)
    aux = double(imgRGGB(i1-1:i1+1,i2-1:i2+1)).*M2;
    imgRGB3(i1, i2, 1) = uint8(sum(aux(:))/sum(M2(:)));
    imgRGB3(i1, i2, 2) = imgRGGB(i1,i2);
    aux = double(imgRGGB(i1-1:i1+1,i2-1:i2+1)).*M1;
    imgRGB3(i1, i2, 3) = uint8(sum(aux(:))/sum(M1(:)));
otherwise %Impar(lv)-impar(lu)
    imgRGB3(i1, i2, 1) = imgRGGB(i1,i2);
    aux = double(imgRGGB(i1-1:i1+1,i2-1:i2+1)).*M4;
    imgRGB3(i1, i2, 2) = uint8(sum(aux(:))/sum(M4(:)));
    aux = double(imgRGGB(i1-1:i1+1,i2-1:i2+1)).*M3;
    imgRGB3(i1, i2, 3) = uint8(sum(aux(:))/sum(M3(:)));
end
end
end
```

B ₁₁	G ₁₂	B ₁₃	G ₁₄	B ₁₅	G ₁₆
G ₂₁	R ₂₂	G ₂₃	R ₂₄	G ₂₅	R ₂₆
B ₃₁	G ₃₂	B ₃₃	G ₃₄	B ₃₅	G ₃₆
G ₄₁	R ₄₂	G ₄₃	R ₄₄	G ₄₅	R ₄₆
B ₅₁	G ₅₂	B ₅₃	G ₅₄	B ₅₅	G ₅₆
G ₆₁	R ₆₂	G ₆₃	R ₆₄	G ₆₅	R ₆₆

$$I(u) = [u \quad 1] \times \begin{bmatrix} a_1 \\ a_0 \end{bmatrix}$$

$$\begin{bmatrix} I(u) \\ I(u+2) \end{bmatrix} = \begin{bmatrix} u & 1 \\ u+2 & 1 \end{bmatrix} \times \begin{bmatrix} a_1 \\ a_0 \end{bmatrix}$$

$$\begin{bmatrix} I(u) \\ I(u+2) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix} \times \begin{bmatrix} a_1 \\ a_0 \end{bmatrix}$$

$$\begin{bmatrix} a_1 \\ a_0 \end{bmatrix} = \begin{bmatrix} -0.5 & 0.5 \\ 1 & 0 \end{bmatrix} \times \begin{bmatrix} I(u) \\ I(u+2) \end{bmatrix}$$

$$I(u+1) = [1 \quad 1] \times \left(\begin{bmatrix} -0.5 & 0.5 \\ 1 & 0 \end{bmatrix} \times \begin{bmatrix} I(u) \\ I(u+2) \end{bmatrix} \right)$$

$$I(u+1) = [1 \quad 1] \times \begin{bmatrix} -0.5I(u) + 0.5I(u+2) \\ I(u) \end{bmatrix}$$

$$I(u+1) = \frac{I(u) + I(u+2)}{2}$$

$$I(v+1) = \frac{I(v) + I(v+2)}{2}$$

B ₁₁	G ₁₂	B ₁₃	G ₁₄	B ₁₅	G ₁₆
G ₂₁	R ₂₂	G ₂₃	R ₂₄	G ₂₅	R ₂₆
B ₃₁	G ₃₂	B ₃₃	G ₃₄	B ₃₅	G ₃₆
G ₄₁	R ₄₂	G ₄₃	R ₄₄	G ₄₅	R ₄₆
B ₅₁	G ₅₂	B ₅₃	G ₅₄	B ₅₅	G ₅₆
G ₆₁	R ₆₂	G ₆₃	R ₆₄	G ₆₅	R ₆₆

$$\begin{bmatrix} a_2 \\ a_1 \\ a_0 \end{bmatrix} = \begin{bmatrix} 1/8 & -1/4 & 1/8 \\ -3/4 & 1 & -1/4 \\ 1 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} I(u) \\ I(u+2) \\ I(u+4) \end{bmatrix}$$

$$I(u+1) = [1 \quad 1 \quad 1] \times \left(\begin{bmatrix} 1/8 & -1/4 & 1/8 \\ -3/4 & 1 & -1/4 \\ 1 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} I(u) \\ I(u+2) \\ I(u+4) \end{bmatrix} \right)$$

$$I(u) = [u^2 \quad u \quad 1] \times \begin{bmatrix} a_2 \\ a_1 \\ a_0 \end{bmatrix} \quad I(u+1) = [1 \quad 1 \quad 1] \times \begin{bmatrix} \frac{I(u) - 2I(u+2) + I(u+4)}{8} \\ -3I(u) + 4I(u+2) - I(u+4) \\ 4I(u) \end{bmatrix}$$

$$\begin{bmatrix} I(u) \\ I(u+2) \\ I(u+4) \end{bmatrix} = \begin{bmatrix} u^2 & u & 1 \\ (u+2)^2 & u+2 & 1 \\ (u+4)^2 & u+4 & 1 \end{bmatrix} \times \begin{bmatrix} a_2 \\ a_1 \\ a_0 \end{bmatrix}$$

$$\begin{bmatrix} I(u) \\ I(u+2) \\ I(u+4) \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 4 & 2 & 1 \\ 16 & 4 & 1 \end{bmatrix} \times \begin{bmatrix} a_2 \\ a_1 \\ a_0 \end{bmatrix}$$

$$I(u+1) = \frac{3I(u) + 6I(u+2) - I(u+4)}{8}$$

$$I(u+3) = \frac{-I(u) + 6I(u+2) + 3I(u+4)}{8}$$


```
%% Bicuadrada
imgRGB4 = zeros(lv, lu, 3, 'uint8');
vp = [3/8; 3/4; -1/8];
for i1=2:lv-3
    for i2=2:lu-3
        switch bitor(mod(i1,2)*2,mod(i2,2))
            case 0 % Par(lv)-par(lu)
                imgRGB4(i1, i2, 2) = uint8(double(imgRGGB(i1,i2+(-1:2:3))))*vp);
                imgRGB4(i1, i2, 3) = imgRGGB(i1, i2);
            case 1 % Par(lv)-impar(lu)
                imgRGB4(i1, i2, 2) = imgRGGB(i1, i2);
                imgRGB4(i1, i2, 3) = uint8(double(imgRGGB(i1,i2+(-1:2:3))))*vp);
            case 2 % Impar(lv)-par(lu)
                imgRGB4(i1, i2, 1) = uint8(double(imgRGGB(i1,i2+(-1:2:3))))*vp);
                imgRGB4(i1, i2, 2) = imgRGGB(i1, i2);
            otherwise %Impar(lv)-impar(lu)
                imgRGB4(i1, i2, 1) = imgRGGB(i1, i2);
                imgRGB4(i1, i2, 2) = uint8(double(imgRGGB(i1,i2+(-1:2:3))))*vp);
        end
    end
end
end
```

```
for i1=2:lv-3
    for i2=2:lu-3
        if(mod(i1,2))
            imgRGB4(i1, i2, 3) =
uint8(vp'*double(imgRGB4(i1+(-1:2:3),i2))));
        else
            imgRGB4(i1, i2, 1) =
uint8(vp'*double(imgRGB4(i1+(-1:2:3),i2))));
        end
    end
end
```

$$I(u) = [u^3 \quad u^2 \quad u \quad 1] \times \begin{bmatrix} a_3 \\ a_2 \\ a_1 \\ a_0 \end{bmatrix}$$

$$\begin{bmatrix} I(u) \\ I(u+2) \\ I(u+4) \\ I(u+6) \end{bmatrix} = \begin{bmatrix} u^3 & u^2 & u & 1 \\ (u+2)^3 & (u+2)^2 & u+2 & 1 \\ (u+4)^3 & (u+4)^2 & u+4 & 1 \\ (u+6)^3 & (u+6)^2 & u+6 & 1 \end{bmatrix} \times \begin{bmatrix} a_3 \\ a_2 \\ a_1 \\ a_0 \end{bmatrix}$$

$$\begin{bmatrix} I(u) \\ I(u+2) \\ I(u+4) \\ I(u+6) \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 8 & 4 & 2 & 1 \\ 64 & 16 & 4 & 1 \\ 216 & 36 & 6 & 1 \end{bmatrix} \times \begin{bmatrix} a_3 \\ a_2 \\ a_1 \\ a_0 \end{bmatrix}$$

$$\begin{bmatrix} a_3 \\ a_2 \\ a_1 \\ a_0 \end{bmatrix} = \begin{bmatrix} -1/48 & 1/16 & -1/16 & 1/48 \\ 1/4 & -5/8 & 1/2 & -1/8 \\ -11/12 & 3/2 & -3/4 & 1/6 \\ 1 & 0 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} I(u) \\ I(u+2) \\ I(u+4) \\ I(u+6) \end{bmatrix}$$

$$\begin{bmatrix} a_3 \\ a_2 \\ a_1 \\ a_0 \end{bmatrix} = \begin{bmatrix} -1/48 & 1/16 & -1/16 & 1/48 \\ 1/4 & -5/8 & 1/2 & -1/8 \\ -11/12 & 3/2 & -3/4 & 1/6 \\ 1 & 0 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} I(u) \\ I(u+2) \\ I(u+4) \\ I(u+6) \end{bmatrix}$$

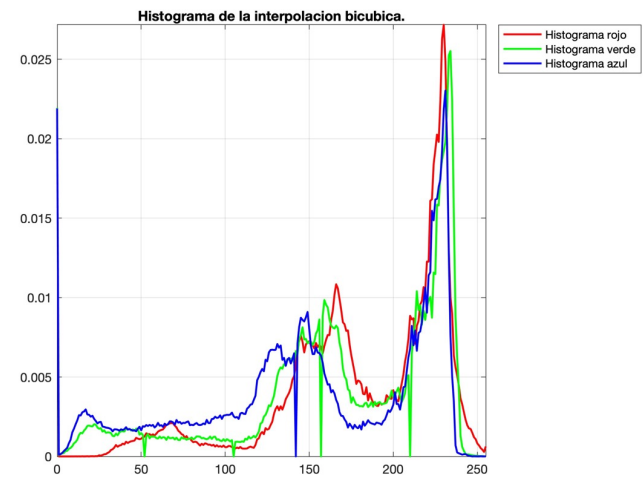
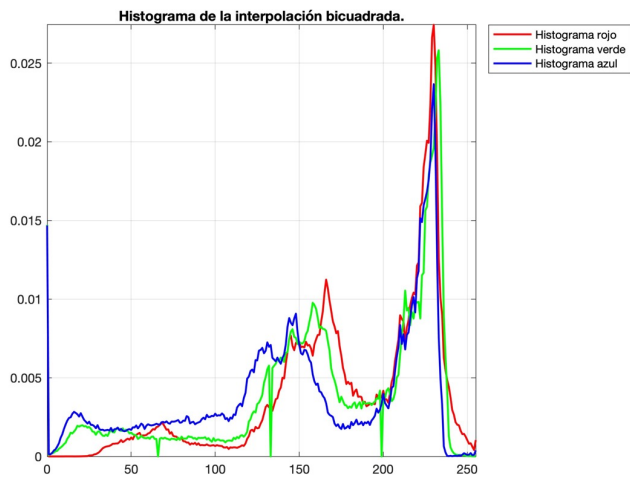
$$I(u+3) = [27 \quad 9 \quad 3 \quad 1] \times \left(\begin{bmatrix} -1/48 & 1/16 & -1/16 & 1/48 \\ 1/4 & -5/8 & 1/2 & -1/8 \\ -11/12 & 3/2 & -3/4 & 1/6 \\ 1 & 0 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} I(u) \\ I(u+2) \\ I(u+4) \\ I(u+6) \end{bmatrix} \right)$$

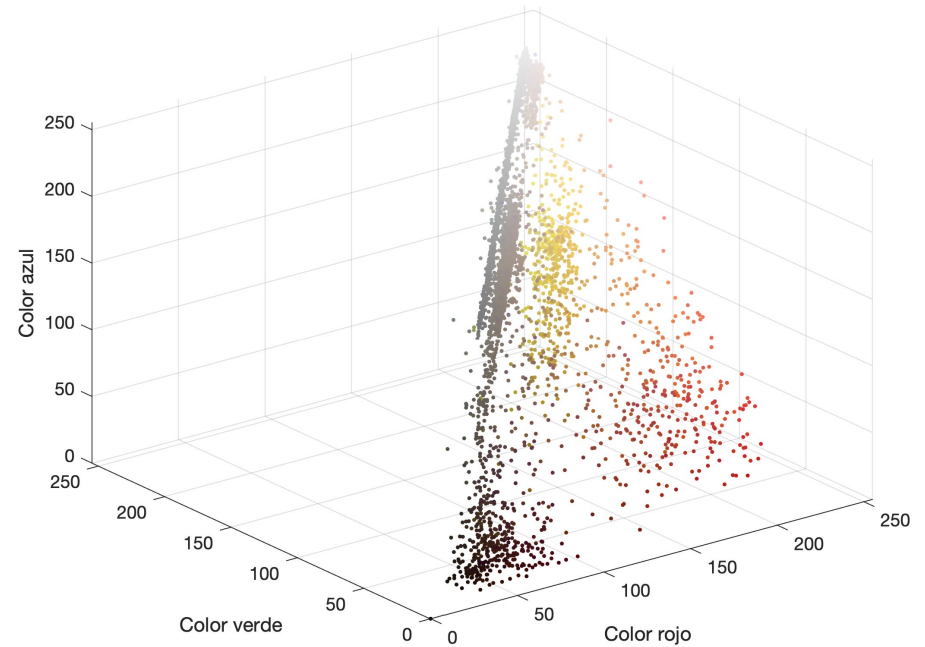
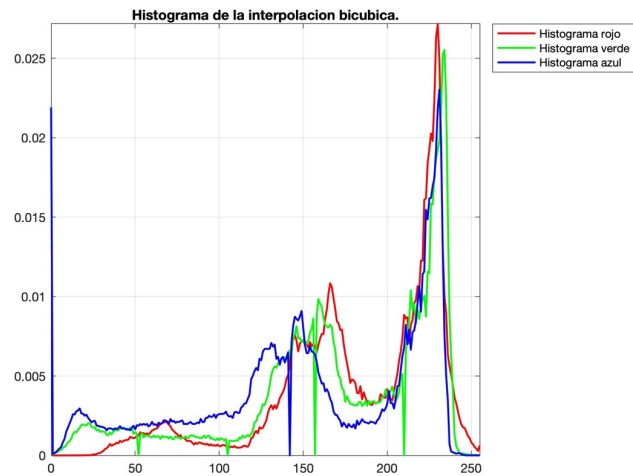
$$I(u+3) = [27 \quad 9 \quad 3 \quad 1] \times \begin{bmatrix} \frac{-I(u) + 3I(u+2) - 3I(u+4) + I(u+6)}{48} \\ \frac{2I(u) - 5I(u+2) + 4I(u+4) - I(u+6)}{8} \\ \frac{-11I(u) + 18I(u+2) - 9I(u+4) + 2I(u+6)}{12} \\ I(u) \end{bmatrix}$$

$$I(u+3) = \frac{-I(u) + 9I(u+2) + 9I(u+4) - I(u+6)}{16}$$

```
imgRGB5 = zeros(lv, lu, 3, 'uint8');
vp = [-1/16; 9/16; 9/16; -1/16];
for i1=4:lv-3
    for i2=4:lu-3
        switch bitor(mod(i1,2)*2,mod(i2,2))
            case 0 % Par(lv)-par(lu)
                imgRGB5(i1, i2, 2) = uint8(double(imgRGGB(i1,i2+(-3:2:3))))*vp);
                imgRGB5(i1, i2, 3) = imgRGGB(i1, i2);
            case 1 % Par(lv)-impar(lu)
                imgRGB5(i1, i2, 2) = imgRGGB(i1, i2);
                imgRGB5(i1, i2, 3) = uint8(double(imgRGGB(i1,i2+(-3:2:3))))*vp);
            case 2 % Impar(lv)-par(lu)
                imgRGB5(i1, i2, 1) = uint8(double(imgRGGB(i1,i2+(-3:2:3))))*vp);
                imgRGB5(i1, i2, 2) = imgRGGB(i1, i2);
            otherwise %Impar(lv)-impar(lu)
                imgRGB5(i1, i2, 1) = imgRGGB(i1, i2);
                imgRGB5(i1, i2, 2) = uint8(double(imgRGGB(i1,i2+(-3:2:3))))*vp);
        end
    end
end
end
```

```
for i1=4:lv-3
    for i2=4:lu-3
        if(mod(i1,2))
            imgRGB5(i1, i2, 3) = uint8(vp'*double(imgRGB5(i1+(-3:2:3),i2,3)));
        else
            imgRGB5(i1, i2, 1) = uint8(vp'*double(imgRGB5(i1+(-3:2:3),i2,1)));
        end
    end
end
end
```





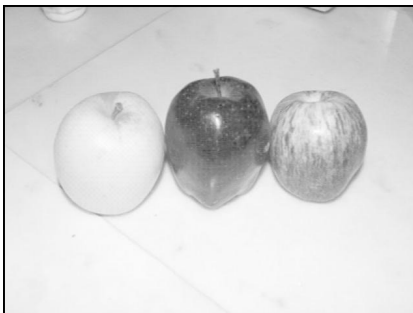
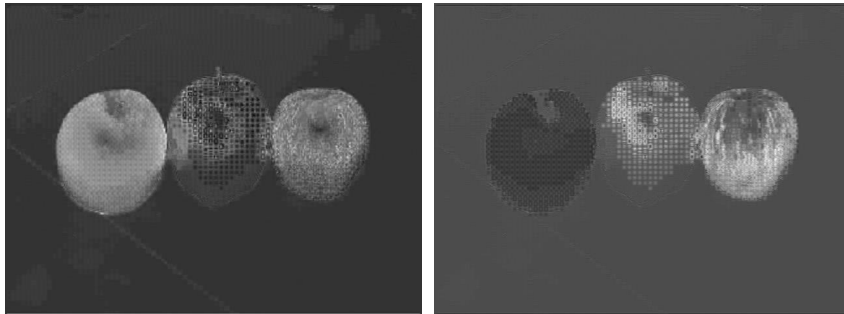
$$\begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = \begin{bmatrix} 0.412453 & 0.357580 & 0.180423 \\ 0.212671 & 0.715160 & 0.072169 \\ 0.019334 & 0.119193 & 0.950227 \end{bmatrix} \begin{pmatrix} R \\ G \\ B \end{pmatrix}$$

$$(0.950456, 1.000000, 1.088754)$$

$$L^* = 116 \left[f \left(\frac{Y}{Y_w} \right) \right] - 16$$

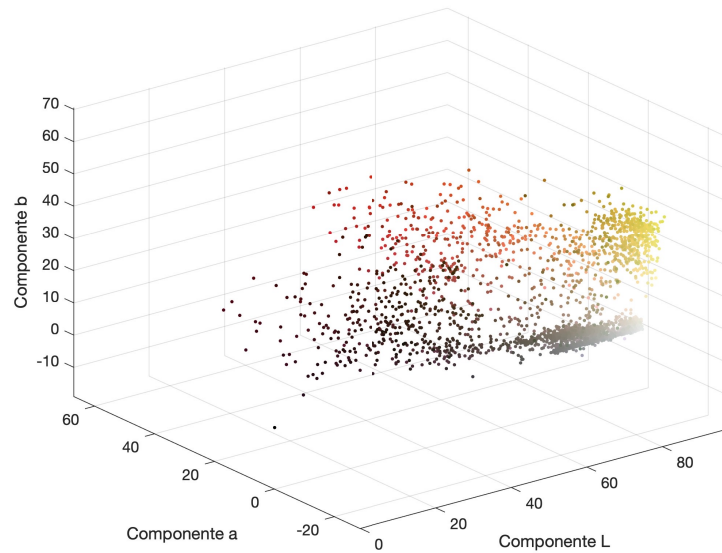
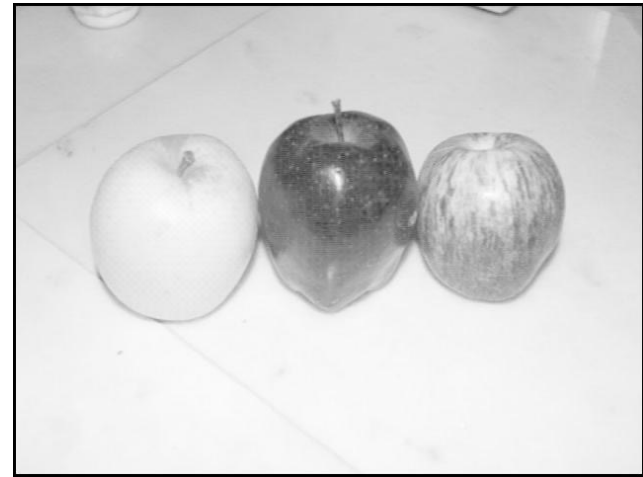
$$a = 500 \left[f \left(\frac{X}{X_w} \right) - f \left(\frac{Y}{Y_w} \right) \right]$$

$$b = 200 \left[f \left(\frac{Y}{Y_w} \right) - f \left(\frac{Z}{Z_w} \right) \right]$$



$$f(t) = \begin{cases} t^{\frac{1}{3}} & \text{si } t > \left(\frac{6}{29}\right)^3 \\ \frac{1}{3} \left(\frac{29}{6}\right)^2 t + \frac{4}{29} & \text{si } t \leq \left(\frac{6}{29}\right)^3 \end{cases}$$

```
M_RGB_XYZ = [ 0.412453 0.357580 0.180423; ...  
              0.212671 0.715160 0.072169; ...  
              0.019334 0.119193 0.950227];  
XYZ_w = [0.950456; 1; 1.088754];  
V_RGV = V_color'/255;  
V_XYZ = M_RGB_XYZ*V_RGV;  
V_XYZ_w = V_XYZ./(XYZ_w*ones(1,lu*lv));  
V_ft = ((1/3)*(29/6)^2)*V_XYZ_w+(4/29);  
ind_ft = (V_XYZ_w>(6/29)^3);  
V_ft(ind_ft) = V_XYZ_w(ind_ft).^(1/3);  
V_Lab = zeros(3, lv*lu);  
V_Lab(1,:) = 116*V_ft(2,:)-16;  
V_Lab(2,:) = 500*(V_ft(1,:)-V_ft(2,:));  
V_Lab(3,:) = 200*(V_ft(2,:)-V_ft(3,:));  
imgL = zeros(lv, lu, 1);  
imga = zeros(lv, lu, 1);  
imgb = zeros(lv, lu, 1);  
imgL(:) = V_Lab(1,:);  
imga(:) = V_Lab(2,:);  
imgb(:) = V_Lab(3,:);  
imgLab = zeros(lv, lu, 3);  
imgLab(:, :, 1) = imgL;  
imgLab(:, :, 2) = imga;  
imgLab(:, :, 3) = imgb;
```



Tinte:

$$h = \tan^{-1} \left(\frac{b}{a} \right)$$

Saturación:

$$C = \sqrt{a^2 + b^2}$$

Binarización

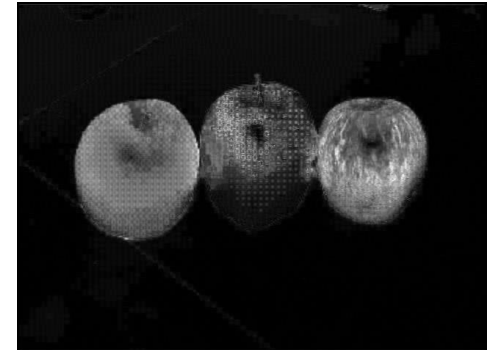
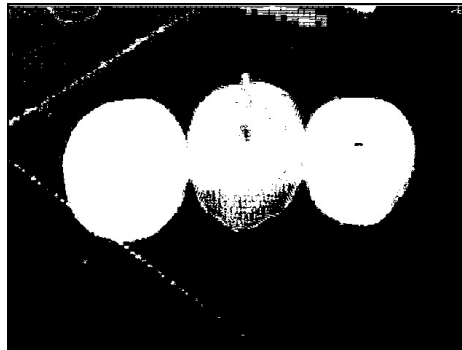
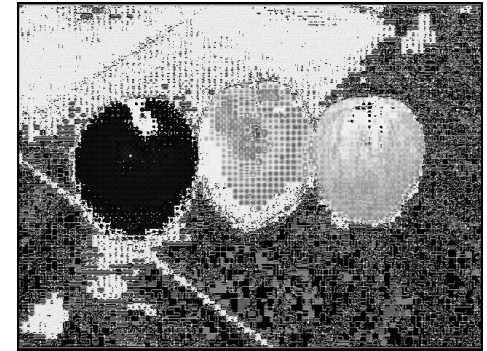
$$M_1 = C > Q_3(C)$$

Apertura

$$M_2 = M_1 \odot D_1 = (M_1 \ominus D_1) \oplus D_1$$

Cierre

$$M_3 = M_2 \odot D_2 = (M_2 \oplus D_2) \ominus D_2$$



```
imgH = atan(imgb./imga);  
imgC = sqrt(imga.^2+imgb.^2);
```

```
imgM = imgC>quantile(imgC(:),.75);  
imgS = zeros(lv,lu,3,'uint8');  
imgS2 = zeros(lv,lu,3,'uint8');
```

```
imgS_C = zeros(lv, lu);  
imgS_C(imgM) = imgR(imgM);  
imgS(:, :, 1) = imgS_C;
```

```
imgS_C = zeros(lv, lu);  
imgS_C(imgM) = imgG(imgM);  
imgS(:, :, 2) = imgS_C;
```

```
imgS_C = zeros(lv, lu);  
imgS_C(imgM) = imgB(imgM);  
imgS(:, :, 3) = imgS_C;
```

```
se1 = strel('disk',10);  
se2 = strel('disk',7);  
imgM1 = imopen(imgM, se1);  
imgM2 = imclose(imgM1, se2);
```

```
imgS_C = zeros(lv, lu);  
imgS_C(imgM2) = imgR(imgM2);  
imgS2(:, :, 1) = imgS_C;
```

```
imgS_C = zeros(lv, lu);  
imgS_C(imgM2) = imgG(imgM2);  
imgS2(:, :, 2) = imgS_C;
```

```
imgS_C = zeros(lv, lu);  
imgS_C(imgM2) = imgB(imgM2);  
imgS2(:, :, 3) = imgS_C;
```

Grandeur

$$\Delta E_{ab} = \sqrt{\Delta L^2 + \Delta a^2 + \Delta b^2}$$

$$\Delta L = L - L(v_i, u_i)$$

Binarización

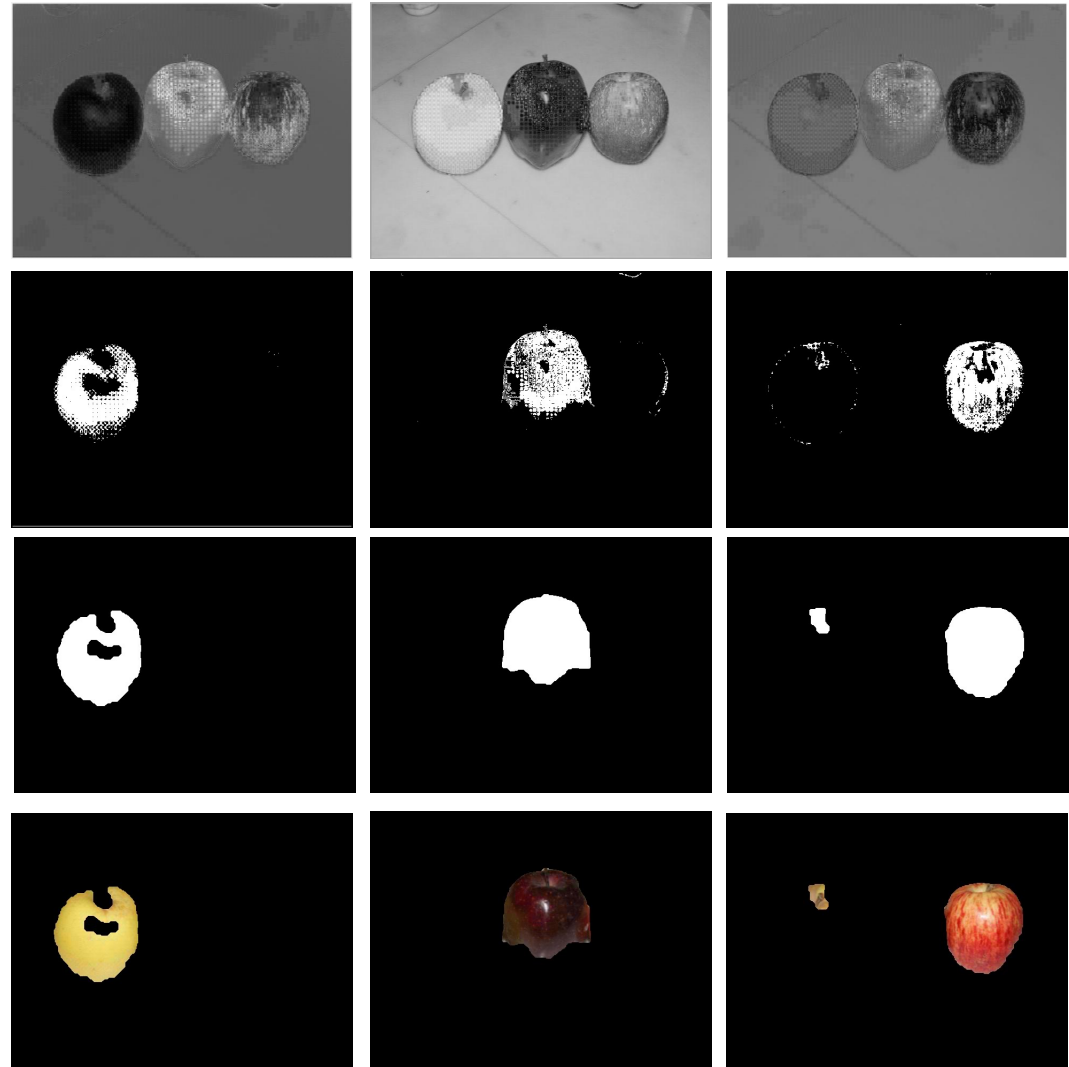
$$M_1 = \Delta E_{ab} < P_5(C)$$

Cierre

$$M_2 = M_1 \odot D_1 = (M_1 \oplus D_1) \ominus D_1$$

Apertura

$$M_3 = M_2 \odot D_2 = (M_2 \ominus D_2) \oplus D_2$$




```
[u_p,v_p] = getpts(f2);
u_p = floor(u_p);
v_p = floor(v_p);

n_p = length(u_p);
for i1=1:n_p
    imgGD = sqrt((imgL-imgL(v_p(i1),u_p(i1))).^2+...
                (imga-imga(v_p(i1),u_p(i1))).^2+...
                (imgb-imgb(v_p(i1),u_p(i1))).^2);
    imgGDM = imgGD<quantile(imgGD(:),.05);
    imgGDM1 = imclose(imgGDM, se1);
    imgGDM2 = imopen(imgGDM1, se2);
    imgGDS = zeros(lv,lu,3,'uint8');

    imgS_C = zeros(lv, lu);
    imgS_C(imgGDM2) = imgR(imgGDM2);
    imgGDS(:, :, 1) = imgS_C;

    imgS_C = zeros(lv, lu);
    imgS_C(imgGDM2) = imgG(imgGDM2);
    imgGDS(:, :, 2) = imgS_C;

    imgS_C = zeros(lv, lu);
    imgS_C(imgGDM2) = imgB(imgGDM2);
    imgGDS(:, :, 3) = imgS_C;
end
```

Grandeur

$$\Delta E_{ab} = \sqrt{\Delta L^2 + \Delta a^2 + \Delta b^2}$$

$$\Delta L = L - L(v_i, u_i)$$

Histograma del grandeur

$$h_{\Delta E_{ab}}(i) = \#\{\Delta E_{ab}(v, u) = i\}$$

$$N_{\Delta E_{ab}} = \max_i (h_{\Delta E_{ab}}(i))$$

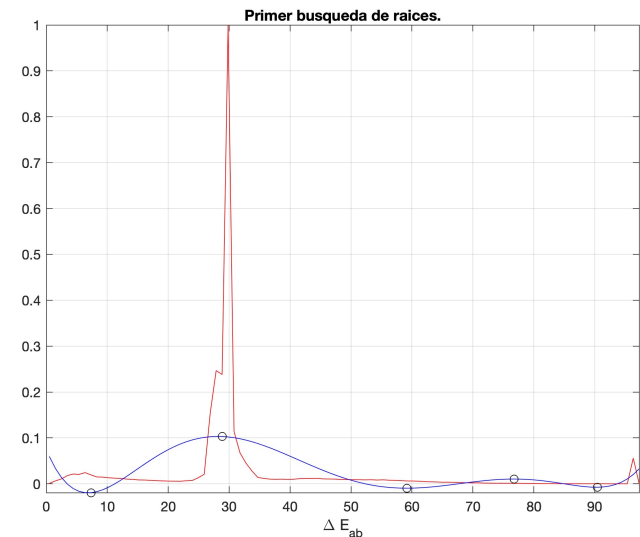
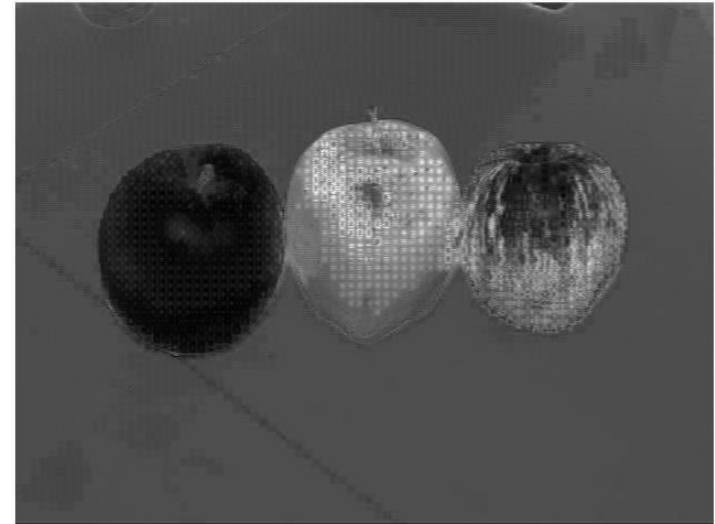
Normalización el histograma

$$P_{\Delta E_{ab}}(i) = \frac{h_{\Delta E_{ab}}(i)}{N_{\Delta E_{ab}}}$$

Regresión polinomial de la normalización

$$f_6(\Delta E_{ab}) = [a_6 \quad a_5 \quad a_4 \quad a_3 \quad a_2 \quad a_1 \quad a_0] \times \begin{bmatrix} \Delta E_{ab}^6 \\ \Delta E_{ab}^5 \\ \Delta E_{ab}^4 \\ \Delta E_{ab}^3 \\ \Delta E_{ab}^2 \\ \Delta E_{ab} \\ 1 \end{bmatrix}$$

$$f_6(\Delta E_{ab}(v_i, u_i)) \cong P_{\Delta E_{ab}}(v_i, u_i)$$



Regresión polinomial de la normalización

$$\begin{bmatrix} \vdots \\ P_{\Delta E_{ab}(v_{i-1}, u_{i-1})} \\ P_{\Delta E_{ab}(v_i, u_i)} \\ P_{\Delta E_{ab}(v_{i+1}, u_{i+1})} \\ \vdots \end{bmatrix} = [a_6 \quad a_5 \quad a_4 \quad a_3 \quad a_2 \quad a_1 \quad a_0] \times \begin{bmatrix} \Delta E_{ab}(v_{i-1}, u_{i-1})^6 & \Delta E_{ab}(v_i, u_i)^6 & \Delta E_{ab}(v_{i+1}, u_{i+1})^6 & \vdots \\ \cdots & \Delta E_{ab}(v_{i-1}, u_{i-1})^5 & \Delta E_{ab}(v_i, u_i)^5 & \Delta E_{ab}(v_{i+1}, u_{i+1})^5 & \cdots \\ \cdots & \Delta E_{ab}(v_{i-1}, u_{i-1})^4 & \Delta E_{ab}(v_i, u_i)^4 & \Delta E_{ab}(v_{i+1}, u_{i+1})^4 & \cdots \\ \cdots & \Delta E_{ab}(v_{i-1}, u_{i-1})^3 & \Delta E_{ab}(v_i, u_i)^3 & \Delta E_{ab}(v_{i+1}, u_{i+1})^3 & \cdots \\ \cdots & \Delta E_{ab}(v_{i-1}, u_{i-1})^2 & \Delta E_{ab}(v_i, u_i)^2 & \Delta E_{ab}(v_{i+1}, u_{i+1})^2 & \cdots \\ \cdots & \Delta E_{ab}(v_{i-1}, u_{i-1}) & \Delta E_{ab}(v_i, u_i) & \Delta E_{ab}(v_{i+1}, u_{i+1}) & \cdots \\ \cdots & 1 & 1 & 1 & \cdots \end{bmatrix}$$

$$[a_6 \quad a_5 \quad a_4 \quad a_3 \quad a_2 \quad a_1 \quad a_0] = \begin{bmatrix} \Delta E_{ab}(v_{i-1}, u_{i-1})^6 & \Delta E_{ab}(v_i, u_i)^6 & \Delta E_{ab}(v_{i+1}, u_{i+1})^6 & \vdots \\ \cdots & \Delta E_{ab}(v_{i-1}, u_{i-1})^5 & \Delta E_{ab}(v_i, u_i)^5 & \Delta E_{ab}(v_{i+1}, u_{i+1})^5 & \cdots \\ \cdots & \Delta E_{ab}(v_{i-1}, u_{i-1})^4 & \Delta E_{ab}(v_i, u_i)^4 & \Delta E_{ab}(v_{i+1}, u_{i+1})^4 & \cdots \\ \cdots & \Delta E_{ab}(v_{i-1}, u_{i-1})^3 & \Delta E_{ab}(v_i, u_i)^3 & \Delta E_{ab}(v_{i+1}, u_{i+1})^3 & \cdots \\ \cdots & \Delta E_{ab}(v_{i-1}, u_{i-1})^2 & \Delta E_{ab}(v_i, u_i)^2 & \Delta E_{ab}(v_{i+1}, u_{i+1})^2 & \cdots \\ \cdots & \Delta E_{ab}(v_{i-1}, u_{i-1}) & \Delta E_{ab}(v_i, u_i) & \Delta E_{ab}(v_{i+1}, u_{i+1}) & \cdots \\ \cdots & 1 & 1 & 1 & \cdots \end{bmatrix} \setminus \begin{bmatrix} \vdots \\ P_{\Delta E_{ab}(v_{i-1}, u_{i-1})} \\ P_{\Delta E_{ab}(v_i, u_i)} \\ P_{\Delta E_{ab}(v_{i+1}, u_{i+1})} \\ \vdots \end{bmatrix}$$

$$f_6(\Delta E_{ab}) = a_6 \Delta E_{ab}^6 + a_5 \Delta E_{ab}^5 + a_4 \Delta E_{ab}^4 + a_3 \Delta E_{ab}^3 + a_2 \Delta E_{ab}^2 + a_1 \Delta E_{ab} + a_0$$

Derivada de la regresión polinomial

$$g_5(\Delta E_{ab}) = \frac{df_6(\Delta E_{ab})}{d\Delta E_{ab}} = 6a_6 \Delta E_{ab}^5 + 5a_5 \Delta E_{ab}^4 + 4a_4 \Delta E_{ab}^3 + 3a_3 \Delta E_{ab}^2 + 2a_2 \Delta E_{ab} + a_1$$

Derivada de la regresión polinomial

$$g_5(\Delta E_{ab}) = \frac{df_6(\Delta E_{ab})}{d\Delta E_{ab}} = 6a_6\Delta E_{ab}^5 + 5a_5\Delta E_{ab}^4 + 4a_4\Delta E_{ab}^3 + 3a_3\Delta E_{ab}^2 + 2a_2\Delta E_{ab} + a_1$$

Puntos de inflexión en la regresión polinomial

$$Z_{g_5(\Delta E_{ab})} = \{z_i | g_5(z_i) = 0, i = \{1,2,3,4,5\}\}$$

Maximización de la regresión polinomial

$$th_1 = \max_i(f_6(z_i))$$

Selección de región de interés en el histograma normalizado

$$P_{th_1}(j) = P_{\Delta E_{ab}}(i) \leftrightarrow i < th_1$$

Regresión polinomial de región de interés

$$h_6(\Delta E_{ab}) = b_6\Delta E_{ab}^6 + b_5\Delta E_{ab}^5 + b_4\Delta E_{ab}^4 + b_3\Delta E_{ab}^3 + b_2\Delta E_{ab}^2 + b_1\Delta E_{ab} + b_0$$

$$[b_6 \quad b_5 \quad b_4 \quad b_3 \quad b_2 \quad b_1 \quad b_0] = \begin{bmatrix} \Delta E_{ab}(v_{i-1}, u_{i-1})^6 & \Delta E_{ab}(v_i, u_i)^6 & \Delta E_{ab}(v_{i+1}, u_{i+1})^6 & \cdots \\ \cdots & \Delta E_{ab}(v_{i-1}, u_{i-1})^5 & \Delta E_{ab}(v_i, u_i)^5 & \Delta E_{ab}(v_{i+1}, u_{i+1})^5 & \cdots \\ \cdots & \Delta E_{ab}(v_{i-1}, u_{i-1})^4 & \Delta E_{ab}(v_i, u_i)^4 & \Delta E_{ab}(v_{i+1}, u_{i+1})^4 & \cdots \\ \cdots & \Delta E_{ab}(v_{i-1}, u_{i-1})^3 & \Delta E_{ab}(v_i, u_i)^3 & \Delta E_{ab}(v_{i+1}, u_{i+1})^3 & \cdots \\ \cdots & \Delta E_{ab}(v_{i-1}, u_{i-1})^2 & \Delta E_{ab}(v_i, u_i)^2 & \Delta E_{ab}(v_{i+1}, u_{i+1})^2 & \cdots \\ \cdots & \Delta E_{ab}(v_{i-1}, u_{i-1}) & \Delta E_{ab}(v_i, u_i) & \Delta E_{ab}(v_{i+1}, u_{i+1}) & \cdots \\ & 1 & 1 & 1 & \cdots \end{bmatrix} \setminus \begin{bmatrix} \vdots \\ P_{th_1}(v_{i-1}, u_{i-1}) \\ P_{th_1}(v_i, u_i) \\ P_{th_1}(v_{i+1}, u_{i+1}) \\ \vdots \end{bmatrix}$$

Regresión polinomial de región de interés

$$h_6(\Delta E_{ab}) = b_6 \Delta E_{ab}^6 + b_5 \Delta E_{ab}^5 + b_4 \Delta E_{ab}^4 + b_3 \Delta E_{ab}^3 + b_2 \Delta E_{ab}^2 + b_1 \Delta E_{ab} + b_0$$

Derivada de la regresión polinomial de la región de interés

$$h'_6(\Delta E_{ab}) = 6b_6 \Delta E_{ab}^5 + 5b_5 \Delta E_{ab}^4 + 4b_4 \Delta E_{ab}^3 + 3b_3 \Delta E_{ab}^2 + 2b_2 \Delta E_{ab} + b_1$$

Puntos de inflexión en la regresión polinomial de la región

$$Z_{h'_6(\Delta E_{ab})} = \{r_i | h'_6(r_i) = 0, i = \{1, 2, 3, 4, 5\}\}$$

Validación de los puntos de inflexión

$$Z_{val} = \{p_k = r_i | \mu_i - 3\sigma_i < r_i < \mu_i + 3\sigma_i\}$$

$$\mu_i = \sum_{\substack{j=1 \\ j \neq i}}^N \frac{r_j}{N} \quad \sigma_i = \sqrt{\frac{s_i}{N}} \quad s_i = \sum_{\substack{j=1 \\ j \neq i}}^N (r_j - \mu_j)^2$$

Selección de umbral

$$th_2 = \max_k(Z_{val})$$



Imagen RGB

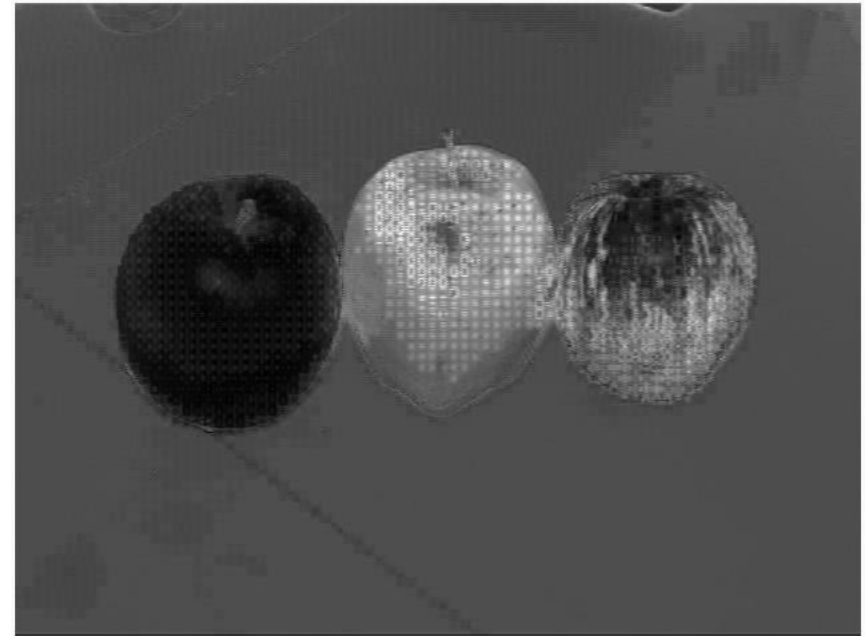
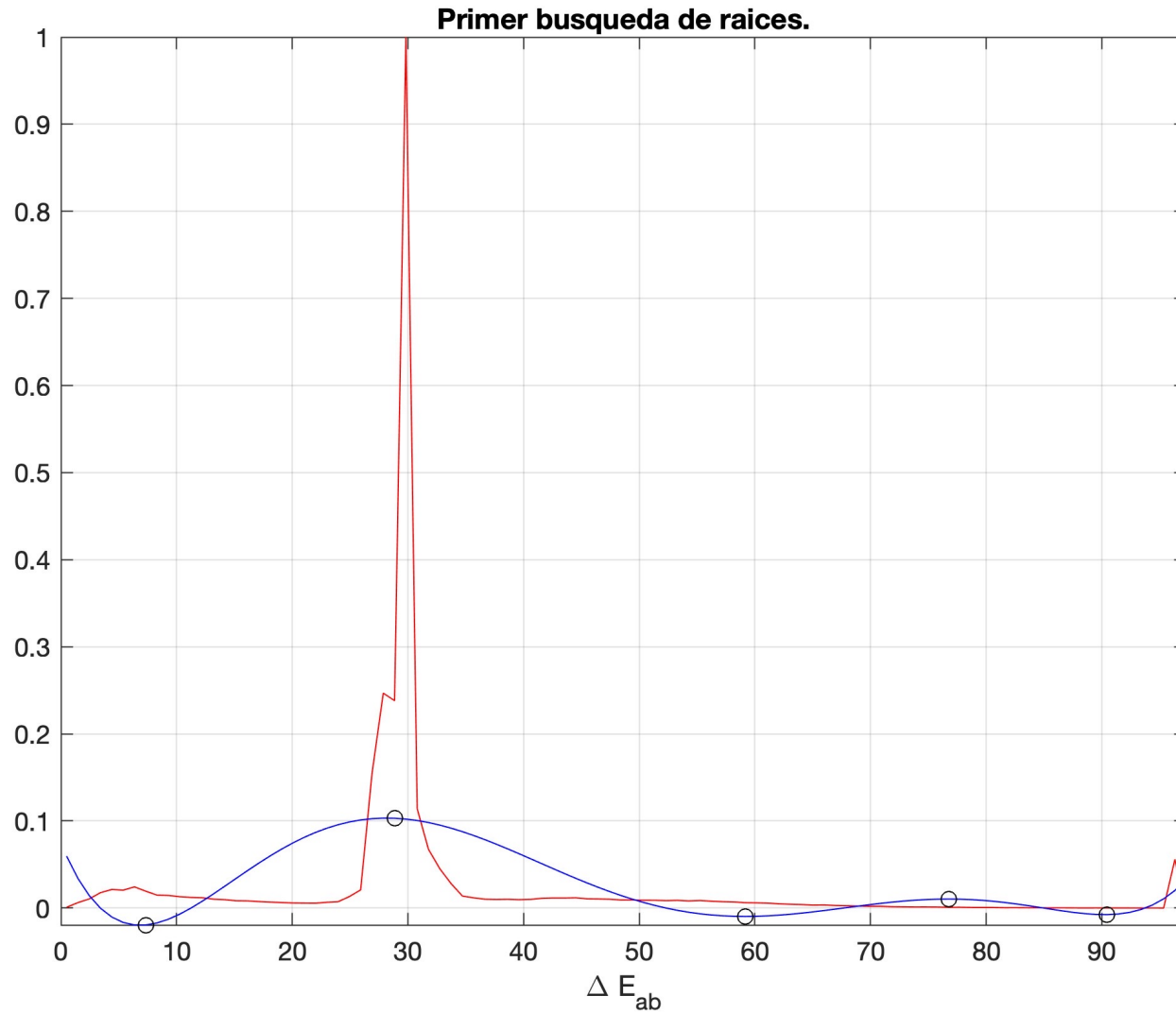
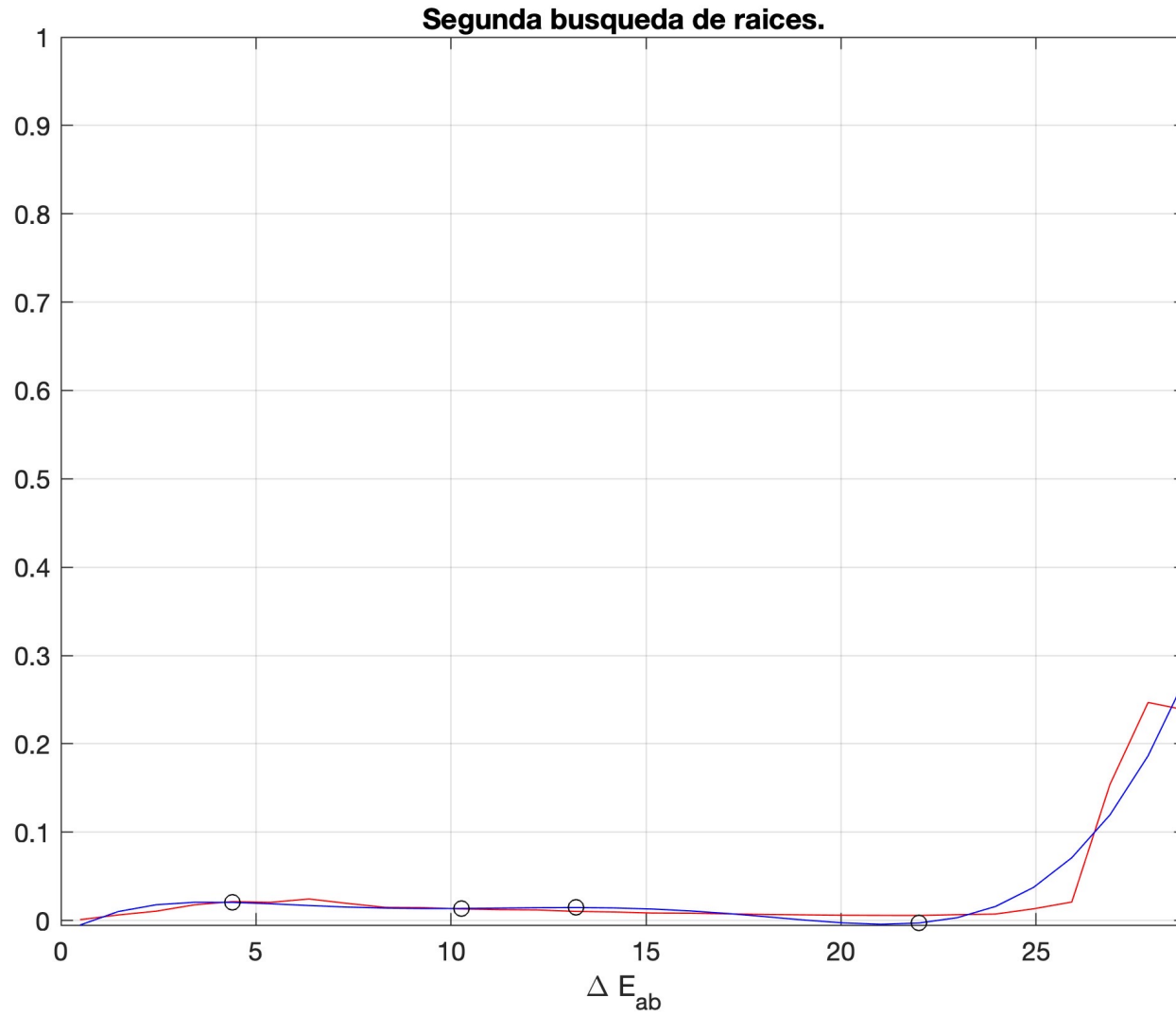
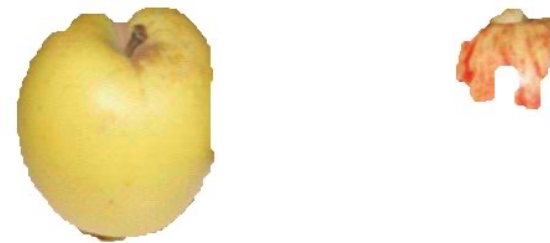
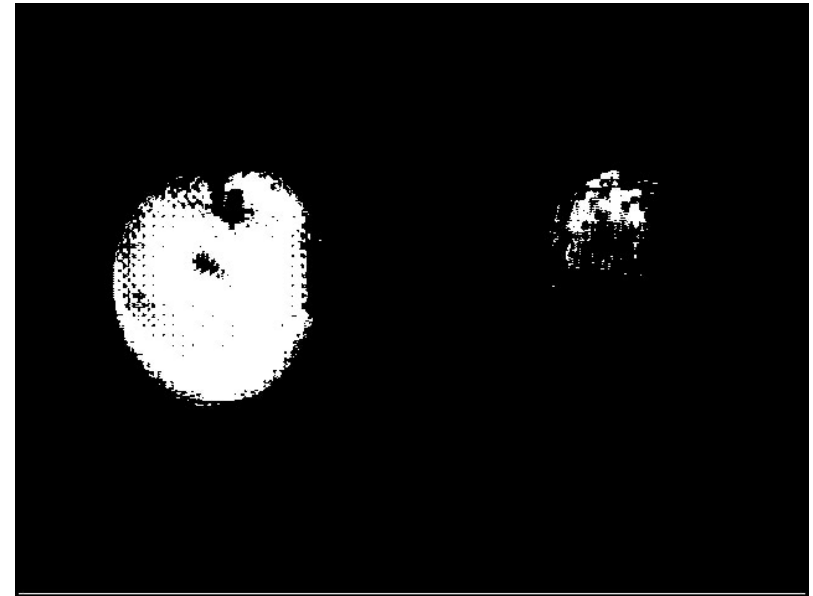
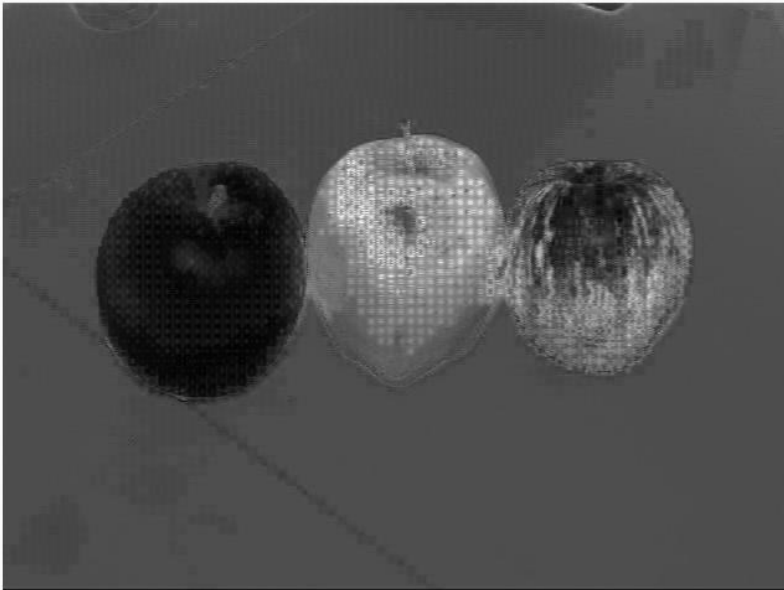
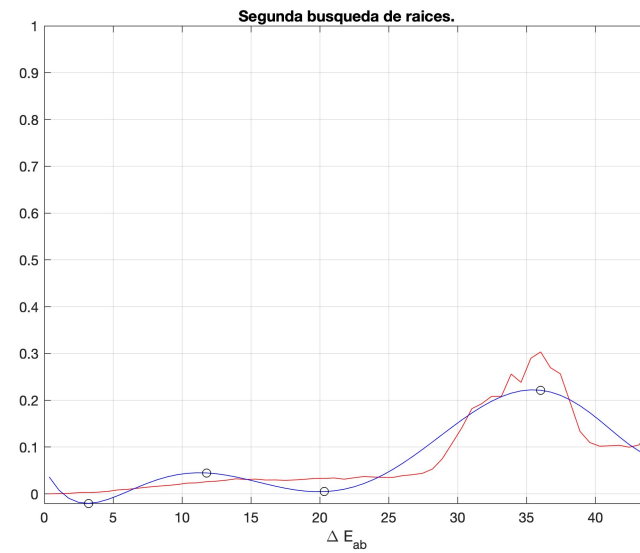
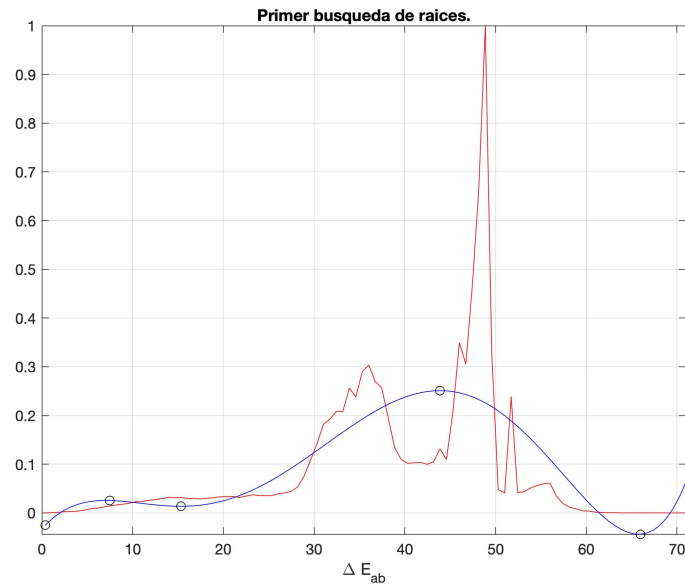
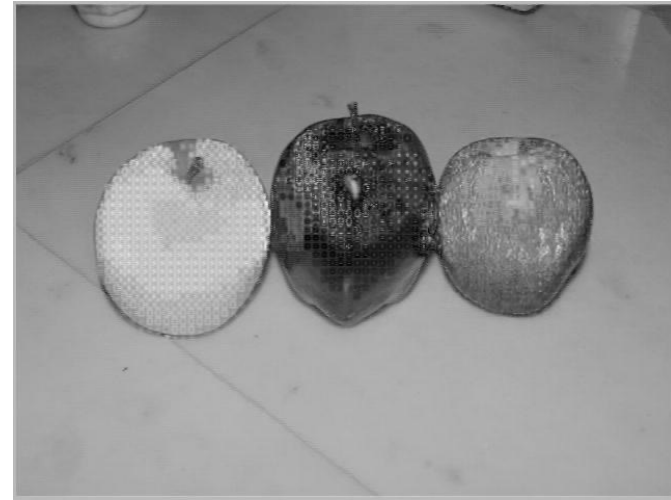


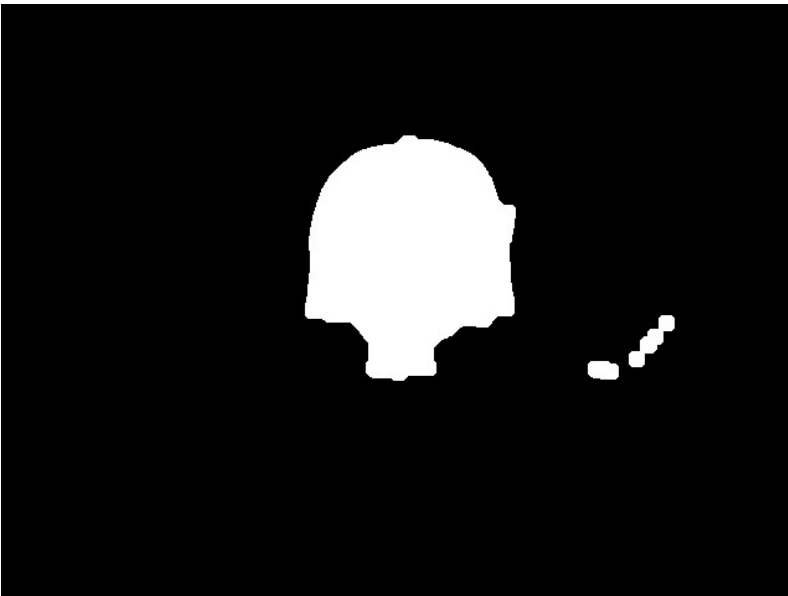
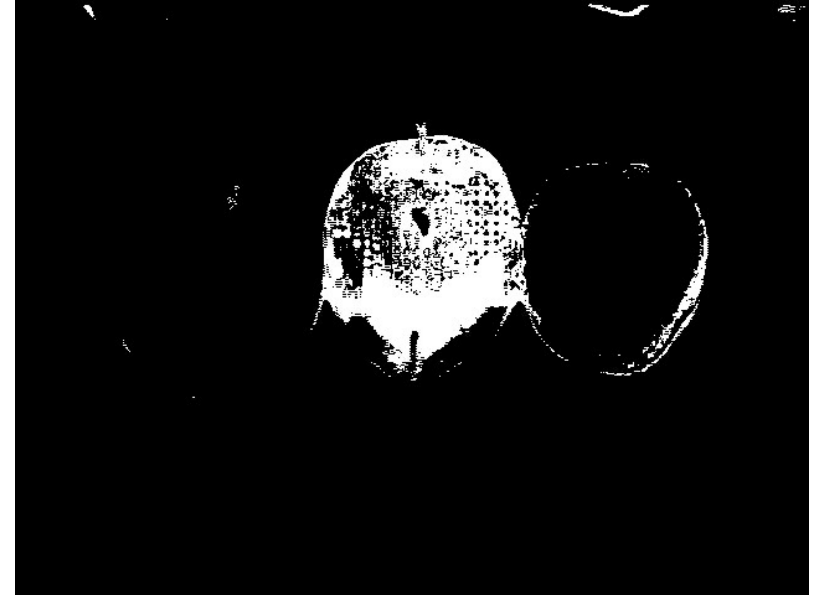
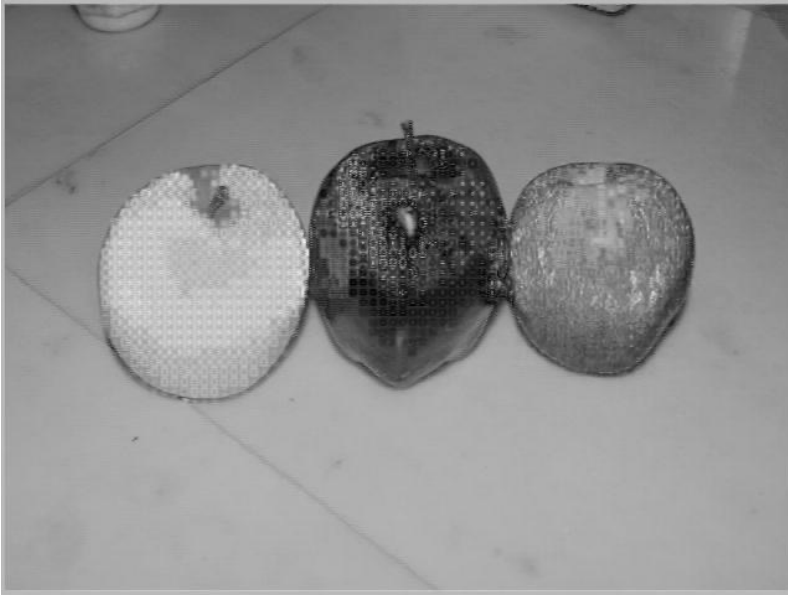
Imagen grandeur ΔE_{ab}

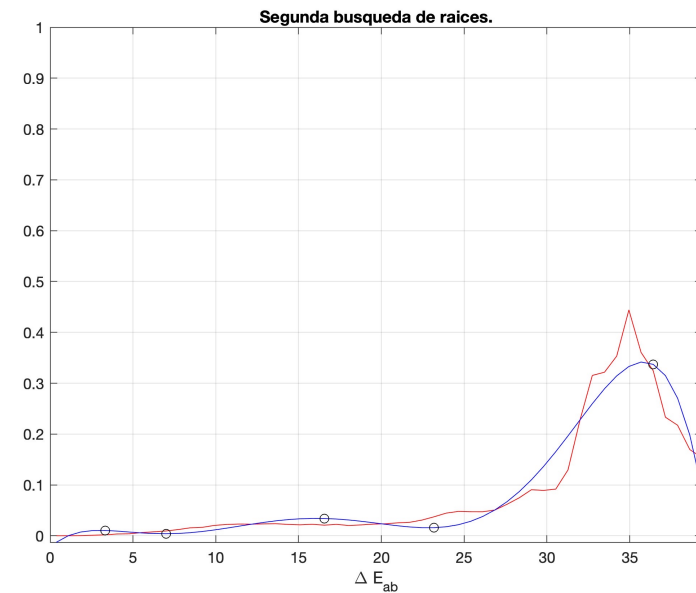
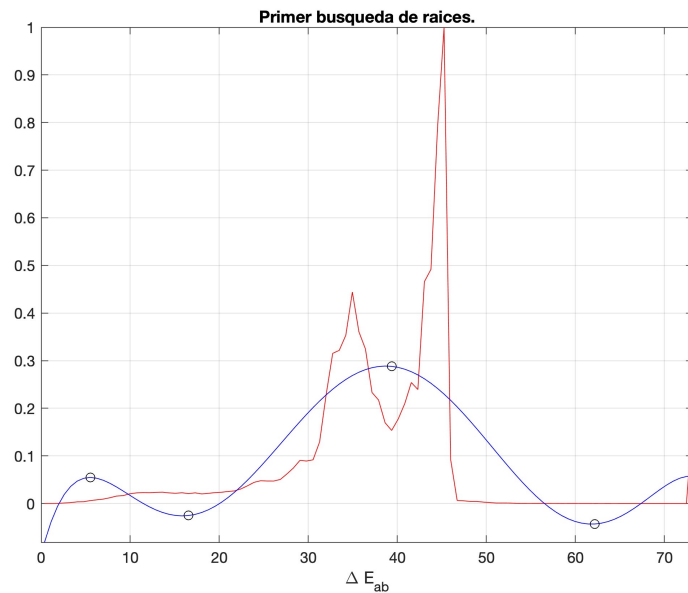
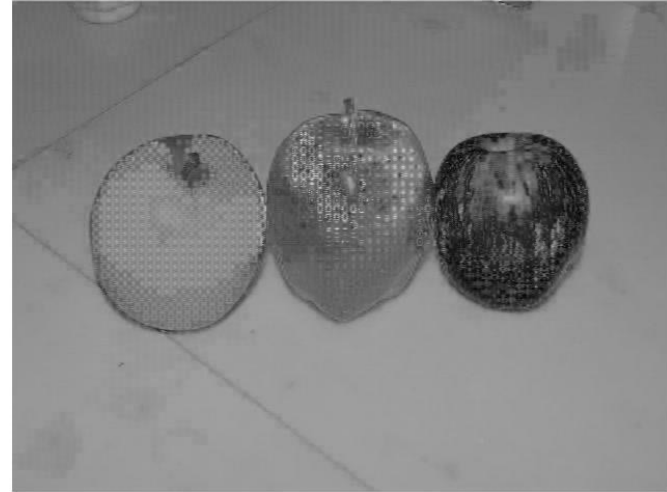


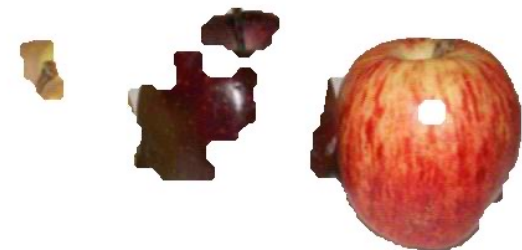
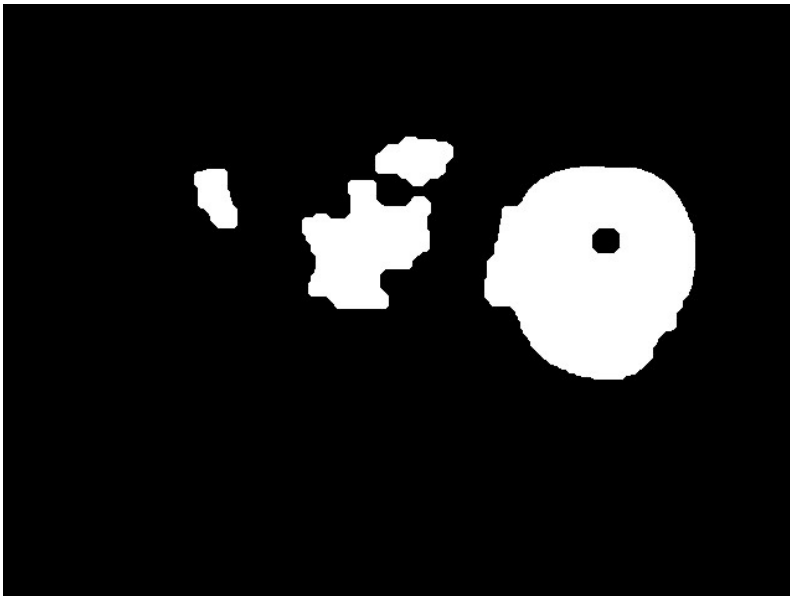
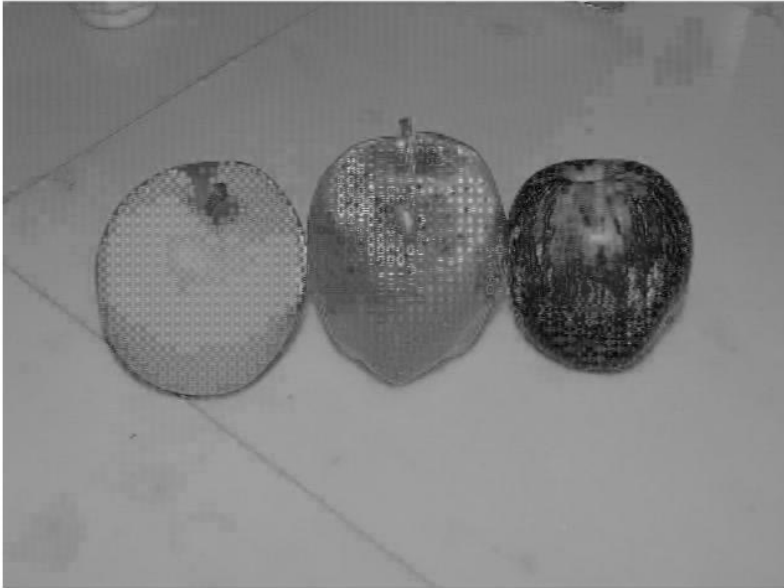












```
[hGD, nGD] = hist(imgGD(:),Np);
hGD_N = hGD/max(hGD);
vC = ((nGD'*ones(1,Ng+1)).^(ones(Np,1)*(0:Ng)))\hGD_N';
fx = ((nGD'*ones(1,Ng+1)).^(ones(Np,1)*(0:Ng)))*vC;
vCd = vC(2:Ng+1).*(1:Ng)';
rCd = roots(vCd(Ng:-1:1));

th1 = rCd(1);
idth1 = find(nGD>th1,1,'first');
if isempty(idth1)
    idth1 = Np;
end
mxfx = fx(idth1);
for i2 = 2:Ng-1
    th2 = rCd(i2);
    idth2 = find(nGD>th2,1,'first');
    vlfx = fx(idth2);
    if vlfx>mxfx
        mxfx = vlfx;
        th1 = th2;
        idth1 = idth2;
    end
end
end
```

```
vC2 = ((nGD(1:idth1)'*ones(1,Ng+1)).^(ones(idth1,1)*(0:Ng)))\hGD_N(1:idth1)';
fx2 = ((nGD(1:idth1)'*ones(1,Ng+1)).^(ones(idth1,1)*(0:Ng)))*vC2;
vC2d = vC2(2:Ng+1).*(1:Ng)';
rC2d = roots(vC2d(Ng:-1:1));

id_rC2d_v = (rC2d>0)&(rC2d<th1);
rC2d_v = rC2d(id_rC2d_v);
n_rC2d_v = length(rC2d_v);
id_rC2d_v = zeros(n_rC2d_v,1);
for i2=1:n_rC2d_v
    id_rC2d_v(i2) = find(nGD>rC2d_v(i2),1,'first');
end
v_rC2d_v = fx2(id_rC2d_v);
st_rC2d = zeros(n_rC2d_v,2);
for i2=1:n_rC2d_v
    st_rC2d(i2, 1) = mean([v_rC2d_v(1:(i2-1));v_rC2d_v((i2+1):n_rC2d_v)]);
    st_rC2d(i2, 2) = std([v_rC2d_v(1:(i2-1));v_rC2d_v((i2+1):n_rC2d_v)]);
end
id_st = ((st_rC2d(:,1)-3*st_rC2d(:, 2))<v_rC2d_v)&((st_rC2d(:,1)+3*st_rC2d(:, 2))>v_rC2d_v);
id_st_v = prod(id_st);
```



```
if id_st_v
    idth = find(nGD>rC2d_v(1),1,'first');
else
    idth = find(nGD>rC2d_v(2),1,'first');
end
```

```
th_p = nGD(idth);
imgGDth = imgGD<th_p;
imgGDthM1 = imclose(imgGDth, se1);
imgGDthM2 = imopen(imgGDthM1, se2);
```

```
imgGDSth = zeros(lv, lu, 3, 'uint8');
```

```
imgS_C = 255*ones(lv, lu);
imgS_C(imgGDthM2) = imgR(imgGDthM2);
imgGDSth(:, :, 1) = imgS_C;
```

```
imgS_C = 255*ones(lv, lu);
imgS_C(imgGDthM2) = imgG(imgGDthM2);
imgGDSth(:, :, 2) = imgS_C;
```

```
imgS_C = 255*ones(lv, lu);
imgS_C(imgGDthM2) = imgB(imgGDthM2);
imgGDSth(:, :, 3) = imgS_C;
```



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